

DOCUMENT PROPOSAL & AGREEMENT

PACK2

Activity 03 — Maintenance Service Simulation · Day 2

Expediente 2 — Propuesta y acuerdo de servicio · Día 2

Case / Caso	CASE 07 — Laboratorio Clínico Vida
Proposal number / Nº de propuesta	PROP-001
Date / Fecha	2026-08-26
Valid until / Vigencia	15 días
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Document 05 — Technical proposal / Lissett Menéndez

Documento 05 — Propuesta técnica

5.1 Problem statement

Laboratorio Clínico Vida is experiencing connection drops between clinical analysers and the internal portal, accompanied by UPS alarms. These issues threaten operations and data integrity right before an upcoming accreditation audit.

El Laboratorio Clínico Vida está experimentando desconexiones recurrentes entre los analizadores clínicos y el portal interno, además de alarmas en el sistema UPS. Estos problemas comprometen la operación y la integridad de los datos justo antes de una auditoría de acreditación.

5.2 Objective of the service

Re-establish network stability between analysers and the portal, replace/service the failing UPS, and document all maintenance logs to meet accreditation standards.

Restablecer la estabilidad de red entre los analizadores y el portal, realizar mantenimiento o reemplazo de baterías del UPS y documentar los registros de mantenimiento para cumplir con la auditoría de acreditación.

5.3 Methodology

#	PHASE	ACTIVITIES IN THIS CASE (ENGLISH)	DELIVERABLE	ESTIMATED TIME
1	Assessment	Scope identification, initial compromise analysis, threat vector evaluation	Reporte de Evaluación Inicial / Matriz de Riesgos	1 - 2 días
2	Preparation	Environment setup, backup creation, isolation of compromised systems, access control staging	Plan de Intervención y Backups de Seguridad	1 día
3	Intervention	Patch management, vulnerability remediation, malware removal, configuration updates	Sistemas Remediados y Configuración Actualizada	2 - 4 días
4	Verification	Penetration testing, vulnerability scanning, continuous log monitoring	Reporte de Auditoría y Verificación de Seguridad	1 - 2 días
5	Closure	Final stakeholder debrief, documentation of lessons learned, project sign-off	Reporte Final de Cierre y Lecciones Aprendidas	1 día

5.4 Tools selected and justification

TOOL	CATEGORY	COST / LICENCE	WHY THIS TOOL FOR THIS CASE (ENGLISH)
Nmap	Network Scanner	Open Source / Free	Used for rapid port discovery and identifying exposed entry points across network subnets.
Wireshark	Packet Analyzer	Open Source / Free	Enables deep packet inspection to analyze malicious network traffic patterns and protocol anomalies.
Nessus	Vulnerability Scanner	Commercial / Subscription	Provides automated scanning to detect known CVEs and system misconfigurations quickly.

Splunk / ELK	SIEM / Log Management	Freemium / Commercial	Centralizes log aggregation to track event timelines and verify threat mitigation across systems.
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5.5 Resources required

TYPE	RESOURCE REQUIRED	QUANTITY	PROVIDED BY (US / CLIENT)
Human	Senior IT Network Specialist / Systems Maintenance Technician	2	Nosotros (In-house IT)
Technical	Managed Ethernet Switch, Diagnostic Multimeter, Wireshark	3	Nosotros (In-house IT)
Material	Cat6 Shielded Ethernet Cables, UPS Replacement Battery Pack	1 kit	Nosotros / Cliente
Time	Ventana de mantenimiento fuera de horas pico del laboratorio	10 horas	Cliente
Information	Protocolos del fabricante, mapa de red, manual del UPS	1 set	Cliente

5.6 Quotation

CONCEPT	QUANTITY	UNIT PRICE (US\$)	SUBTOTAL (US\$)	NOTE
Technician labour	10	\$20.00	\$200.00	Trabajo técnico en sitio (10 hrs)
Materials / spare parts	1	\$180.00	\$180.00	Baterías de UPS y cables Cat6
Transport	1	\$10.00	\$10.00	Movilidad local de personal

Licences / third-party services	0	\$0.00	\$0.00	N/A (Uso de herramientas libres)
Contingency (10 %)	1	\$39.00	\$39.00	10% de imprevistos sobre subtotal
TOTAL	-	-	\$429.00	Presupuesto total del proyecto

5.7 Payment terms and warranty

Internal budget allocation. All replaced parts (UPS batteries, cabling) carry a 12-month warranty. Service includes full documentation for the accreditation audit.

Asignación presupuestaria interna. Todos los repuestos (baterías de UPS, cableado) tienen 12 meses de garantía. El servicio incluye documentación completa para la auditoría de acreditación.

Client approval / Aprobación del cliente:

Name: _____ Signature: _____ Date: _____

Document 06 — Service Level Agreement (SLA) / Valeria Contreras

Documento 06 — Acuerdo de nivel de servicio

Service hours / Horario de servicio	24/7 Operations Support
Maintenance window / Ventana	Saturdays 18:00 – 22:00 (Low sample volume)
Availability committed / Disponibilidad	99.5% uptime for portal and analyser connectivity

6.1 Priority matrix

PRIORITY	DEFINITION FOR THIS CLIENT	RESPONSE TIME	RESOLUTION TIME
P1 Critical	Analyser disconnects entirely or portal goes down during lab operating hours	≤ 30 min	≤ 2 hours
P2 High	UPS alarm active or intermittent latency on portal response.	≤ 2 hours	≤ 6 hours
P3 Medium	Single workstation disconnected from lab portal; non-critical log warnings	≤ 4 hours	≤ 24 hours
P4 Low	Schedule routine preventive maintenance or audit log export request.	≤ 12 hours	≤ 48 hours

6.2 Services included

Network infrastructure maintenance for analysers, UPS power backup management, portal database maintenance, and audit compliance reporting.

Mantenimiento de infraestructura de red para analizadores, gestión del sistema UPS, mantenimiento de base de datos del portal e informes de cumplimiento para la auditoría.

6.3 Exclusions

Internal hardware failures inside proprietary clinical analysers (requires vendor specialist).

Fallas internas de hardware en analizadores clínicos propietarios (requiere especialista del fabricante).

6.4 Escalation path

Level 1: On-duty IT Technician → Level 2: Lead Systems Administrator → Level 3: IT & Operations Director.

Nivel 1: Técnico de TI de turno → Nivel 2: Administrador de Sistemas → Nivel 3: Director de TI y Operaciones.

6.5 Consequences of non-compliance

Failure to meet P1 resolution times will result in internal service quality penalties and mandatory reporting to management.

Si no se cumplen los tiempos de solución para problemas críticos (P1), se aplicarán sanciones internas en la evaluación de calidad de TI y se enviará un informe obligatorio a la gerencia.

For the provider / Por el proveedor

For the client / Por el cliente

Document 07 — Risk matrix and rollback plan / Jazmín Peña

Documento 07 — Matriz de riesgos y plan de reversión

ID	RISK	PROBABILITY	IMPACT	PREVENTIVE MEASURE	CONTINGENCY / ROLLBACK
R-01	Total power outage while replacing the UPS battery	Low	High	Connect clinical analysers to a backup power line before starting work.	Switch back to the main power line immediately.
R-02	Portal database failure or data loss	Low	Critical	Take a full offline backup of the database before network changes.	Restore the database from the verified backup copy.
R-03	Configuration mismatch on the new network switch	Medium	High	Pre-configure IP ranges and VLAN settings before mounting the switch.	Reconnect the original switch using its previous configuration.
R-04	Maintenance logs rejected during the accreditation audit	Low	High	Confirm log formats with the auditor prior to finalizing reports.	Re-format report files to match the exact audit template.

R-05	Network cable damage during rack cleanup	Low	Medium	Use brand-new, pre-tested Cat6 shielded patch cords.	Replace any damaged cable segment with a spare cord right away.
R-06	IP assignment mismatch between new switch and analysers	Low	High	Record all static IP bindings for each analyser prior to cutover.	Revert switch port settings back to their original state.

7.1 Rollback procedure

If maintenance fails, immediately reconnect network cables to the original switch, restore the previous IP routing settings, turn on the primary UPS line, and test analyser communication.

Si el mantenimiento falla, reconectar inmediatamente los cables de red al switch original, restaurar la configuración IP previa, encender la línea principal del UPS y probar la conexión de los analizadores.

7.2 Backup and verification

A complete offline database backup will be saved to an external encrypted drive, and restoration will be verified by executing a test query on a secondary server before starting maintenance.

Se realizará un respaldo completo en frío de la base de datos en un disco externo cifrado, y se probará la restauración mediante una consulta de prueba en un servidor secundario antes de iniciar el trabajo.

Document 08 — Work order and maintenance Schedule/ Marjory Lima

Documento 08 — Orden de trabajo y calendario

Work order number / Nº de orden	WO-2026-001
Scheduled date and window / Fecha y ventana	}wednesday, Oct 26, 2026 7:00 – 10:15
Authorised by / Autoriza	Dra. María Elena Torres (Directora de Operaciones)

8.1 Tasks authorised

#	TASK TO BE PERFORMED (ENGLISH)	TOOL / METHOD	EST. TIME	RESPONSIBLE
1	Take complete offline cold backup of portal database	PostgreSQL Dump / USB Drive	30 min	Lead Systems Admin
2	Switch clinical analysers to auxiliary power line	Manual Transfer Switch	15 min	Maintenance Tech
3	Replace UPS internal battery pack and run self-test	Multimeter / Manufacturer Tool	45 min	Maintenance Tech
4	Pre-configure and install main network switch	SSH Console / Console Cable	45 min	Network Specialist
5	Connect Cat6 patch cables and verify VLANs	Cable Tester / Network Analyzer	30 min	Network Specialist
6	Reconnect analysers and verify static IP communication	Ping Test / Vendor Interface	30 min	Network Specialist
7	Perform network speed, latency, and log tests	Wireshark / Nmap	30 min	Lead Systems Admin
8	Document tasks, sign work order, and handover	Maintenance Log / PDF Sign	15 min	IT Team Lead

8.2 Preventive maintenance schedule

RECURRING TASK	FREQUENCY	RESPONSIBLE	TOOL	WHY THIS FREQUENCY?
UPS Battery & Load Test	Monthly	Maintenance Tech	Multimeter / UPS Software	Prevents power cuts during critical lab testing hours.

Analysers Cable Check	Quarterly	Network Specialist	Cable Tester	Movement near machines causes physical wear over time.
Portal DB Optimization	Weekly	DB Administrator	PostgreSQL Scripts	High volume of lab results slows queries and portal response.
Audit Log Review	Monthly	IT Team Lead	Log Analyzer / Backup Utility	Guarantees system logs are ready for accreditation audits.
Network Switch Inspection	Bi-annually	Network Specialist	SNMP Monitoring / Console	Ensures firmware stability and port availability.

Authorisation / Autorización: the client authorises the execution of the tasks listed above within the agreed window, and confirms that a backup of the information has been made or that responsibility for it is accepted.

El cliente autoriza la ejecución de las tareas listadas dentro de la ventana acordada y confirma que se realizó respaldo de la información o que asume la responsabilidad sobre ella.

Name: _____ Signature: _____ Date: _____